LESSON PLAN

HELICOPTER THEORY (11)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	27
Total No. of Lesson Plans	27	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Low speed wind tunnel and wind tunnel balances
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Low speed wind tunnel and wind tunnel balances. Demonstration on pressure plotting experiments. Measurements of lift and drag in wind tunnel.
5. Specific Objectives	11.1.5.1 Trainee will be able to explain about wind tunnel operation

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration on pressure plotting experiments. Measurements of lift and drag in wind tunnel. Safety: Ensure tunnel is off before accessing balance. Instructor supervision required during setup.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a wind tunnel?
(b) Name two types of aerodynamic forces measured in a wind tunnel.
(c) Why is a low speed wind tunnel used for preliminary testing?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Principles of Low Speed Wind Tunnel • Components of Wind Tunnel System • Operation of Wind Tunnel Fan • Flow Straightening Screens • Test Section Characteristics	L & Ds	PPT, IB	15 min
(b)	• Types of Wind Tunnel Balances • Strain Gauge Balance Principles • Force Measurement Mechanisms • Moment Measurement Mechanisms • Calibration of Balances	L & Ds	PPT, PDF	20 min
(c)	• Pressure Plotting Experiments • Data Recording Procedures • Lift and Drag Calculation • Analysis of Test Results • Error Sources in Measurement	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how a strain gauge balance measures lift force.
(b) Describe the role of the test section in a low speed wind tunnel.
(c) List three precautions required during wind tunnel balance calibration.

12. Home Assignment and Follow Up	Prepare a brief note on the differences between a low speed wind tunnel and a high speed wind tunnel, referencing the next lesson plan (LP 02) on High Speed Wind Tunnels.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	End of subject	Refer to the master syllabus reference list	-

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min